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WhereTheWaterFlows.jl: Hydrological flow routing on digital elevation models in Julia

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Summary

`WhereTheWaterFlows.jl` (`WWF`) is a Julia package for computing water flow paths and drainage basins on digital elevation models (DEMs) or other hydropotential surfaces. Given a DEM, the package determines *where the water flows* across the landscape, accumulating upslope area and delineating catchment basins. Besides this classic, deterministic routing on surface topographies, WWF also includes routines to route water in subglacial settings as well as uncertainty quantification tools based on a Monte Carlo approach.

Key outputs include the flow-direction field, the upslope contributing area or discharge, the depression-filled DEM, a labelled catchment-basin map, and the location of all sinks. Plotting functions for result visualisation are included. The package is registered in the Julia General registry, has a full test suite and complete online documentation.

WWF implements the widely-used D8 single-flow-direction algorithm, which routes water from each grid cell to its lowest-elevation neighbour among the eight surrounding cells, together with a breach-type depression-filling algorithm to handle closed depressions that would otherwise trap flow (O'Callaghan & Mark, 1984). The underlying graph traversal follows the O(n) recursive strategy of Braun & Willett (2013), giving linear scaling with the number of grid cells. WWF performs comparably to or better than existing routing software, as shown in our benchmarks.

Statement of need

Hydrological flow routing on DEMs is a fundamental operation in geosciences: it underpins catchment delineation, runoff modelling, subglacial hydrology, and landscape-evolution studies. Software implementations of flow routing date back to O'Callaghan & Mark (1984) and, nowadays, many implementations exist. However, `WWF` is the first, and to our knowledge, only native Julia package for this task.

WWF is suitable for any domain where DEM-based flow analysis is required. However, the package has been designed with glaciological applications in mind where it has been used in six publications to date, see section Research impact. The glaciological focus required the implementation of glacier specific aspects, such as Shreve potential routing (Shreve, 1972) in the `Subglacially` submodule, as well as the development of uncertainty quantification via Monte Carlo methods (the `Randomly` submodule). The latter aspect is needed in glaciology as uncertainties in glacier bed DEMs and Shreve-potential are large and thus warrant quantification.

State of the field

Several mature tools exist for hydrological flow routing on DEMs, examples include:

- **GRASS GIS** (command line/C/Python/QGIS), the component `r.watershed` implements several water routing and related algorithms. (GRASS-Team et al., 2026)
- **TauDEM** (command line/ArcGIS) provides command-line tools designed for large-scale catchment analysis. (Tesfa et al., 2011)
- **TopoToolbox** (MATLAB/Python) provides widely used functionalities for topographic analysis including flow routing. (Schwanghart & Scherler, 2014)
- **Whitebox Tools** (Rust/Python) provides an extensive geomorphometry toolkit as a Rust library with Python bindings and unofficial Julia bindings. (Lindsay, 2016)
- **RichDEM** (Python/C++) offers a broad set of flow-routing methods. (Barnes et al., 2014)

To our knowledge, none of the existing routing software packages implement specialised subglacial routing functionality or have uncertainty quantification capabilities.

Software design

The central function is `waterflows(dem)`, which accepts a two-dimensional array of elevation values and returns a named tuple of outputs such as upstream area and catchments. Boundary conditions (open or closed edges, fixed sinks) and other options can be specified via keyword arguments. A novel feature of WWF is the feedback-function (via the `feedback_fn` argument of `waterflows`) which is called on each cell before routing its water downstream, enabling feedbacks to occur. This allows for extensive customisations, as demonstrated in the provided examples (see `examples/` directory) on sediment transport and subglacial melt calculations.

For subglacial routing, the main function is `waterflows_subglacial(surfdem, beddem, dx, f)` with surface and bed DEMs, grid spacing and water pressure as flotation fraction as inputs. In the subglacial setting, the `beddem` and `f` input fields have large uncertainties associated with them. This spurred the development of the Monte Carlo based uncertainty quantification in WWF. The main function there is `map_mc(model, sample, reduce!, n)` which runs the `model` function (i.e. the water routing) on `n` samples of surface, bed and flotation fractions fields and reduces the results with `reduce!` (to avoid huge data volumes). The uncertainty in the input fields is modelled using Gaussian Random Fields, which provide spatially correlated random noise. The random fields are generated using a FFT-based method (Räss et al., 2019). However, the software design allows to couple more sopisticated, external geostatistical models to generete the uncertainty fields.

Example

The code and figure below show an example of a drainage network calculation on a small synthetic DEM:

```julia
using WhereTheWaterFlows, CairoMakie
n   = 200
x = y = range(-π, π, length=n)
dem = sin.(x) .* cos.(y') .+ 0.05 * rand(n, n)

out = waterflows(dem)
plt_area(x, y, out.area)
```

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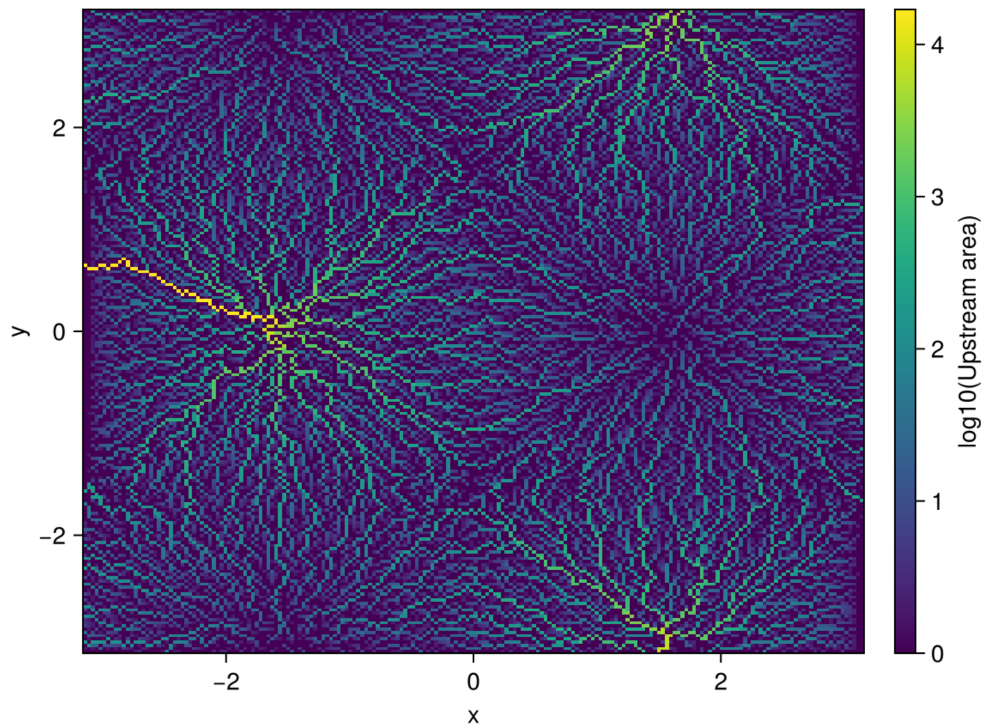

Figure 1: Upslope contributing area on a synthetic DEM from running the provided code example. Brighter colours indicate larger contributing areas, tracing the drainage network emerging from the synthetic topography.

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Research impact

`WhereTheWaterFlows.jl` has been used in published glaciological research: Malczyk et al. (2023) used it to route subglacial melt water and constrain melt fluxes beneath an ice sheet from observations of active subglacial lake recharge. The study employed WWF's uncertainty quantification features. Delaney et al. (2023) built a 2D subglacial sediment transport model on top of WWF. Ogier et al. (2025) employed flow routing to analyse the likelihood of subglacial water pockets in Alpine glaciers, contributing to a new inventory for the Swiss Alps. Horgan et al. (2025) used flow routing to delineate the subglacial catchment of the Kamb Ice Stream. Washam et al. (2026) demonstrated that volcanic heat sources within the Kamb catchment likely contribute to enhanced subglacial melt. Ogier et al. (2026) (in review) used the package to assess the potential contribution of subglacial water reservoirs to the catastrophic 2024 La Bérarde flood.

WWF is used for teaching in the lecture courses "Physics of Glaciers" at ETH Zurich (Switzerland) and "Introduction to geoscientific programming" at Uni Mainz (Germany). The software is archived on Zenodo (Werder et al., 2024) and has accumulated 20+ GitHub stars and 5+ forks since its public release in 2019. It has four contributors and 200+ commits across 20+ tagged releases.

AI usage disclosure

AI tools from Anthropic and OpenAI are used in the development of WWF. Most of the code predates 2022, the advent of capable LLMs, and was thus written by hand. This also applies to the `Randomly` and `Subglacially` submodules which were previously developed in a separate, private repository and only now merged into WWF. Recently, code refactoring was helped by LLMs. Much of the documentation, examples and benchmarks were generated with the help of LLMs. The tools employed to date are Claude Sonnet 4.6 and GPT-5.3-Codex. All AI generated content was reviewed by a human.

The first draft of this manuscript was created by Claude Sonnet 4.6. The tool was used for initial text generation and structural organisation of the paper sections. The text was then reviewed, heavily edited and rewritten by the two authors.

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